## §3.5 Lab Assignment 2 - Quadrature Error Analysis

This lab assignment has three parts. In part I, you will implement some of the quadrature methods described in this chapter and numerically guess how the error appears to depend on value of $h = \frac{b-a}{n}$ for each method. In particular, you will be asked to find whether the data appear to show errors of type $\mathcal{O}(h), \mathcal{O}(h^2), \mathcal{O}(h^3)$, etc.

In part II, you will use Calculus to prove some of the experimentally observed orders.

In part III, you will implement the Adaptive Simpson Rule described in the last section, and compare its efficiency to a Composite Simpson Rule.

### 3.5.1 Part I - Experimental Analysis

In this part you will approximate a test integral whose exact value is known, and use a *log–log* plot to estimate the convergence order from the slope.

Consider

$$I = \int_0^{\pi/2} \sin x \, dx = 1.$$

Let $n$ denote the number of equal subintervals and $h = \frac{b-a}{n}$.

**Question 42.**

*Implement the following five* composite *quadrature rules on* $[0, \pi/2]$*:*

**(a)** *Composite midpoint rule.*

**(b)** *Composite trapezoid rule.*

**(c)** *Composite Simpson's rule.*

**(d)** *Composite Simpson's three-eighths rule.*

**(e)** *Composite Gauss–Legendre rule with* $\varphi_2$ *applied on each subinterval.*

*Use the following sequence as the number of partitions*

$$n = 6 \cdot 2^k \qquad (k = 0, 1, 2, \ldots, K),$$

*so that every method is well-defined for every $n$ (in particular, Simpson requires $n$ even and Simpson* $3/8$ *requires $n$ divisible by* $3$*).*

*For each method and each $n$, compute* $Q_n$ *and the absolute error*

$$E_n := |Q_n - I|.$$

**(a)** *For each method, make a plot of* $\log(E_n)$ *versus* $\log(h)$*. Each plot should clearly show the points corresponding to the different $n$ values.*

**(b)** *From the plot, identify a range of $k$ values where the points lie approximately on a straight line, and visually (or numerically by calculating a rise-over-run) estimate the slope. Record your estimated slope*

*$p$ for each method and interpret it as $E_n = \mathcal{O}(h^p)$.*

**(c)** *Briefly comment on whether the plot shows a plateau for very small $h$ (i.e. error stops decreasing). If so, give a plausible explanation.*

*Use a table to organize your $n, h, E_n, p$ data for each method.*

### 3.5.2 Part II - Theoretical Error Analysis

Our basic approach will be to analyze the 'local' error associated with a single subinterval of a quadrature rule, and then propagate that error across all subintervals to establish a 'global' error relationship on the entire interval of integration $[a,b]$.

All of our quadrature rules so far have used $n$ subdivisions of length $h = \dfrac{b-a}{n}$ of the interval of integration $[a,b]$. A typical subinterval can be represented as $[x_i, x_i+h]$ where $x_i = a+ih$ for $i = 0,1,ldots,n-1$.

**Local Error for Trapezoid Rule**

**Question 43.**

*Recall that the Trapezoid rule uses linear interpolant between successive points. Suppose the linear $f$–interpolant on $[x_i, x_{i+1}]$ is $T_i(x)$. We will define*

$$E_i(h) = \int_{x_i}^{x_i+h} (f(x) - T_i(x))\,dx$$

*Show that*

$$E_i(h) = \int_{x_i}^{x_i+h} \frac{f^{(2)}(\xi)}{2!}(x-x_i)(x-x_i-h)\,dx$$

*for some $\xi \in [x_i, x_i+h]$.*

□

**Warning:** Observe that $\xi$ is not a constant, its value depends on $x$. As such, we can't just pull it outside the integral.

At this point, we will need to use the generalized Mean Value Theorem for Integrals. A proof of this result can be found in a standard Calculus textbook.

**Theorem 3.5.23**

*Let $f$ and $g$ be continuous real functions on the closed interval $[a,b]$ such that:*

$$(\forall x \in [a,b])(g(x) \geq 0) \qquad or \qquad (\forall x \in [a,b])(g(x) \leq 0)$$

*Then there exists a real number $k \in [a,b]$ such that:*

$$\int_a^b f(x)g(x)dx = f(k)\int_a^b g(x)dx$$

Now note that in the interval $[x_i, x_i+h]$, the product $(x-x_i)(x-x_i-h)$ is always negative. Hence using the GMVT for integrals, we can write

$$\mathrm{E}_i(h) = f^{(2)}(\eta)\int_{x_i}^{x_i+h} \frac{(x-x_i)(x-x_i-h)}{2}\,\mathrm{d}x$$

for some $\eta \in [x_i, x_i+h]$.

## Question 44. □

*Use a change of variable to integrate the last integral and show that* $\mathrm{E}_i(h) = \mathcal{O}(h^3)$.

### Global Error for Trapezoid Rule

To propagate local errors across all the subintervals of $[a,b]$, we simply add all the local errors together. Since there are $n = \frac{b-a}{h}$ subintervals, we have the general relationship that

$$\mathrm{E}(h) \approx \mathcal{O}(n\mathrm{E}_i(h)) = \mathcal{O}\left(\frac{b-a}{h}\mathrm{E}_i(h)\right) = \mathcal{O}\left(\frac{\mathrm{E}_i(h)}{h}\right) = \frac{\mathcal{O}(\mathrm{E}_i(h))}{h}$$

which means that we typically lose a power of $h$ going from local to global errors. Notice that the global error $\mathrm{E}(h)$ could be better than this if there happens to be cancellation of local errors.

> **Note:** Make sure your experimental observations in question 42 match the theoretical observations above.

### Simpson's Rule

Let's try the same technique with Simpson's rule. However, this time we need $2n$ subintervals to begin. Suppose $\mathrm{S}_i(x)$ is the quadratic interpolant at the nodes $x_i, x_{i+1}$, and $x_{i+2}$. Then we can write

$$\mathrm{E}_i(h) = \int_{x_i}^{x_{i+2}} (f(x) - \mathrm{S}_i(x))\,\mathrm{d}x = \int_{x_i}^{x_{i+2}} \frac{f^{(3)}(\xi)}{3!}(x-x_i)(x-x_{i+1})(x-x_{i+2})\,\mathrm{d}x$$

Now we run into a problem because the product $(x-x_i)(x-x_{i+1})(x-x_{i+2})$ is not always positive or negative over the interval $[x_i, x_{i+2}]$. In fact, because of symmetry, you can easily check that

$$\int_{x_i}^{x_i+2h} (x-x_i)(x-x_i-h)(x-x_i-2h)\,\mathrm{d}x = 0$$

So clearly, we can't use the GMVT to pull out a $f^{(3)}(\eta)$ term as we did for the Trapezoid rule.

To fix this issue, we will use a clever trick! Suppose we were to ask what the cubic interpolant is through four nodes $x_i, x_{i+1}, x_{i+1}$, and $x_{i+2}$ (i.e., the middle node is counted twice).

## Question 45. □

*Explain why the third order interpolant $\widetilde{S}_i(x)$ through four nodes $x_i, x_{i+1}, x_{i+1}$, and $x_{i+2}$ must be the exact same polynomial as the second order interpolant $S_i(x)$ through three nodes $x_i, x_{i+1}$, and $x_{i+2}$.*

So we replace $S_i(x)$ with $\widetilde{S}_i(x)$ and rewrite

$$E_i(h) = \int_{x_i}^{x_{i+2}} (f(x) - \widetilde{S}_i(x))\, dx = \int_{x_i}^{x_{i+2}} \frac{f^{(4)}(\xi)}{4!}(x - x_i)(x - x_{i+1})(x - x_{i+1})(x - x_{i+2})\, dx$$

and voila! Suddenly, the product is a function of $x$ that is negative everywhere on the interval $[x_i, x_{i+2}]$.

## Question 46.

□

*Use the GMVT for integrals and a change of variables to show that for Simpson's rule $E_i(h) = \mathcal{O}(h^5)$. Consequently, deduce that the global composite Simpson error is typically $\mathcal{O}(h^4)$ on $[a, b]$.*

**General Newton-Cotes**

For the sake of completion, we will include the following analogous result for Newton-Cotes quadrature. A proof can be found at this linked JSTOR article.

Suppose we have $n+1$ equally-spaced interpolation points

$$x_i := a + i\left(\frac{b-a}{n}\right) = a + ih \text{ for } i = 0, 1, \ldots, n,$$

where $n$ represents the degree of the $f$-interpolating polynomial $p_n(x)$. If

$$E(h) = \int_a^b (f(x) - p_n(x))\, dx,$$

then using similar techniques as above, we can show that

- When $n$ is odd, $E(h) = \mathcal{O}(h^{n+2})$.
- When $n$ is even, $E(h) = \mathcal{O}(h^{n+3})$.

### 3.5.3 Part III - Adaptive Quadrature

In this part you will implement an adaptive method and compare how many subintervals it uses to achieve a target accuracy, versus how many *uniform* subintervals are needed by a standard composite rule to reach the same accuracy.

## Question 47.

□

*Use the algorithm in section 3.3 Implement adaptive Simpson quadrature on an interval $[a, b]$ using* `tol` $= 10^{-8}$ *and apply your code to the following integrals on $[0, 1]$:*

$$\int_0^1 \frac{1}{1 + 25(x - \frac{1}{2})^2}\, dx, \qquad \int_0^1 \sin(100x)\, dx, \qquad \int_0^1 \sqrt{x}\, dx.$$

*For each integral:*

- ***(a)*** *Report the adaptive approximation* $\mathrm{Q}_{\text{adapt}}$*, the final number of accepted subintervals m, and the total estimated error. Note that you should be able to find the exact values of each integral by hand.*

- ***(b)*** *Now use a composite Simpson's rule with uniform partition and determine the smallest n such that the approximation satisfies*

$$\left|\mathrm{Q}_n - \int_a^b f(x)\,\mathrm{d}x\right| \leq 10^{-8}.$$

  *Record the smallest n.*

- ***(c)*** *Compare m (adaptive) versus n (uniform). Give a short explanation of what feature(s) of the integrand cause the adaptive method to be more or less effective in each case.*

- ***(d)*** ***Optional:*** *Include a visualization of the final adaptive partition (for example, a plot of subinterval endpoints on* $[0,1]$ *or a histogram of subinterval lengths).*